Numbers That Talk

Divisibility rules and why they work

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Do this with us during the talk. The **why** is on the back.

Warm-up. This year is 2026. Is it divisible by 9? Add the digits: 2 + 0 + 2 + 6 = _____. Is that a multiple of 9? ☐ yes ☐ no. So the remainder of 2026 ÷ 9 is _____.

1 The vanishing digit (Zhiheng)

1. Think of a four-digit number. Not all four digits the same. Don't tell us. **My number:** _____
2. Shuffle its digits into a new number. **Shuffled:** _____
3. Subtract the smaller one from the bigger one. _____ − _____ = _____
4. Circle **one non-zero digit** of the result. Keep it hidden.
5. Read us the other digits, in any order. We name the hidden one. **We said** ____ ☐ right

Example. 5274 → shuffle 7425 → 7425 − 5274 = 2151 → hide the 5 → you read 2, 1, 1 → we say **5**.

2 1001 brings it back (Yueran)

1. Think of a three-digit number. **My number:** _____
2. Write it twice, so it becomes a six-digit number. _____
3. Divide by 7. = _____ (it divides evenly, promise)
4. Divide that by 11. = _____
5. Divide that by 13. = _____ ☐ my number is back

Example. 572 → 572572 → ÷ 7 = 81796 → ÷ 11 = 7436 → ÷ 13 = **572**.

3 Try it after the talk

1. Is 45658 divisible by 37? Cut it into three-digit chunks from the right: 45 | 658. Add: 45 + 658 = _____. Is that a multiple of 37? ☐ yes ☐ no
2. What is 2026 mod 7? Chunks 2 | 026. Alternate: 026 − 2 = _____. That mod 7 is ____, and mod 13 it is ____.
3. 1/7 = 0.142857 142857… Multiply 142857 by 2: _____. By 3: _____. Same digits?

Why does any of this work? → turn over

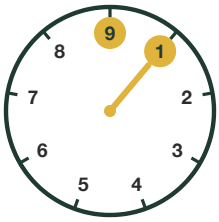
Why it works

Everything below only uses things from class: congruences, Fermat and Euler, and the order of an element.

A. Trick 1 is "10 ≡ 1 (mod 9)"

Theorem. For any n with digits d_0, d_1, … (units first), $n \equiv d_0 + d_1 + \dots \pmod 9$, hence also mod 3.

Proof. $n = d_0 + d_1 \cdot 10 + d_2 \cdot 10^2 + \dots$. Since $10 = 9 + 1$, $10 \equiv 1 \pmod 9$, so $10^i \equiv 1$ and $d_i \cdot 10^i \equiv d_i$. Add up. 3 | 9 gives the mod 3 version. ∎



A clock with only 9 marks. 10 steps land where 1 step lands. So do 100 and 1000.

The trick. Your number n and the shuffled n′ have the same digits, so $n \equiv n' \pmod 9$ and 9 | n − n′. By the theorem the digits of the difference add to a multiple of 9. You read all but one digit x; the sum s you read satisfies $s + x \equiv 0 \pmod 9$, and exactly one x in 1…9 does that. (A 0 is banned because 0 and 9 would look the same.)

2026 → 2+0+2+6 = 10 → 1, so $2026 \equiv 1 \pmod 9$. 7452 → 18, so 9 | 7452.

B. Trick 2 is "1000 ≡ −1 (mod 1001)"

abcabc = abc·1000 + abc = abc·1001, and **1001 = 7 × 11 × 13**. Dividing by 7, 11, 13 in turn is dividing by 1001, so abc comes back. It cannot fail.

More useful: $1000 = 1001 - 1$, so $1000 \equiv -1 \pmod{1001}$. Cut n into three-digit blocks B_0, B_1, B_2, … from the right. Then $n = B_0 + B_1 \cdot 1000 + B_2 \cdot 1000^2 + \dots \equiv B_0 - B_1 + B_2 - \dots \pmod{1001}$, hence mod 7, 11 and 13 at once.

2026 → 026 − 2 = 24 → $24 \equiv 3 \pmod 7$, $\equiv 11 \pmod{13}$, $\equiv 2 \pmod{11}$. All three are the true remainders.

C. One idea, every rule

Block theorem. If $10^k \equiv 1 \pmod d$, then $n \equiv B_0 + B_1 + B_2 + \dots \pmod d$ for k-digit blocks from the right. If $10^k \equiv -1$, then $n \equiv B_0 - B_1 + B_2 - \dots$. Proof: write n in base 10^k and use $(10^k)^j \equiv (\pm 1)^j$, exactly as in A. ∎

d	k	$10^k \equiv$	rule	example
3, 9	1	+1	digit sum	2026 → 10 → 1
11	1	−1	alternate digits	2026 → 6−2+0−2 = 2
101	2	−1	alternate 2-blocks	461267 → 67−12+46 = 101 ✓
27, 37	3	+1	add 3-blocks (999 = 27×37)	45658 → 45+658 = 703 = 19×37 ✓
7, 13, 77	3	−1	alternate 3-blocks	1283945 → 945−283+1 = 663 = 13×51 ✓
41, 271	5	+1	add 5-blocks (99999 = 9×41×271)	$\text{ord}_{41}(10) = 5$

D. Every prime p ≠ 2, 5 gets a rule

Theorem. (1) (p − 1)-digit blocks, added, always work. (2) The shortest *additive* block is $k = \text{ord}_p(10)$, and k | p − 1. (3) If k is even, $10^{k/2} \equiv -1 \pmod p$, so (k/2)-blocks with alternating signs work too.

Proof. (1) p does not divide 10, so Fermat gives $10^{p-1} \equiv 1 \pmod p$; apply the block theorem. (2) By definition of the order, k is the least exponent with $10^k \equiv 1$, and $a^n \equiv 1 \Rightarrow$ ord | n gives k | p − 1. (3) Put $x = 10^{k/2}$. Then $p \mid x^2 - 1 = (x - 1)(x + 1)$. By Euclid's lemma p | x − 1 or p | x + 1; the first contradicts the minimality of k. ∎

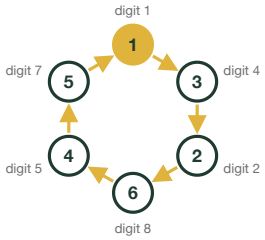
Careful. (3) needs p prime: for d = 33, ord(10) = 2 but 10 is not ≡ −1 (mod 33). For composite d with gcd(d, 10) = 1, Euler gives $10^{\phi(d)} \equiv 1$, so φ(d)-blocks work and the shortest additive block is still $\text{ord}_d(10)$. Example: φ(27) = 18 but $\text{ord}_{27}(10) = 3$, since 999 = 27 × 37.

7: ord = 6 | 6, even, so $10^3 \equiv -1$. 13: ord = 6 | 12. 37: ord = 3 | 36 (odd, no −1 rule). 41: ord = 5 | 40.

E. Block length = period of 1/p

Theorem. If gcd(d, 10) = 1, d > 1, then 1/d is purely repeating with period $\text{ord}_d(10)$.

Proof. Long division starts with remainder $r_0 = 1$ and each step replaces r by 10r mod d, so $r_i \equiv 10^i \pmod d$. The digits repeat from the first i with $r_i = 1$, which is $i = \text{ord}_d(10)$. ∎ Also directly: $10^6/7 - 1/7 = 142857$, so $999999 = 7 \times 142857$.



Remainders of 1 ÷ 7: 1, 3, 2, 6, 4, 5, back to 1. They are 10^i mod 7. Six steps, so six digits.

So "how many digits per block for 7" and "how long is the period of 1/7" are the same question. Both answers are 6.

Bonus. 142857 × 1, 2, …, 6 = 142857, 285714, 428571, 571428, 714285, 857142: the same digits rotated, and × 7 = 999999. Reason: the powers of 10 mod 7 hit every non-zero remainder, so 10 is a *primitive root* mod 7, and k/7 just enters the same cycle at a different point. **Midy:** 142 + 857 = 999 and 076 + 923 = 999 (for 1/13), because $10^3 \equiv -1$.

F. More, if you are curious

Chop-the-last-digit rules from 10^{-1} mod d (7: x − 2y; 13: x + 4y). 2^k and 5^k: last k digits, since $10 \equiv 0$. Coprime factors combine (6 = 2 × 3), non-coprime do not (4 and 6 divide 12, but 24 does not). Other bases: digit sum tests b − 1, alternating sum tests b + 1. ISBN-10 check digits use mod 11 because 11 is prime.